Enrollment Number 241040071029

Total marks: 15 **Data Structures - Quiz 1** **Date: 04/09/2025**

4½

Instructions:

4.5

1. This is a closed book and closed Internet test.
2. MCQs may have multiple correct options.
3. Plagiarism or any form of cheating will lead to disqualification.
4. Use of Internet, Mobile phones, calculators and wearables is not allowed.

Q.1.) Choose the right option. **(0.5 mark)**

int* x, y;

0

A) x is a pointer to a integer, y is a integer
B) y is a pointer to a integer, x is a integer
✓ C) both x and y are pointers to integer types
D) y is a pointer to a integer

Q.2.) What is the primary operation used to add an element to a stack? **(0.5 mark)**

0.5

A) peek B) Enqueue C) Pop ✓ D) Push

Q.3.) What is the principle followed by a Stack data structure? **(0.5 mark)**

0.5

A) FILO B) LILO ✓ C) LIFO D) FIFO E) Random Access

Q.4.) What is the principle followed by a queue data structure? **(0.5 mark)**

0.5

A) FILO B) LILO C) LIFO ✓ D) FIFO E) Random Access

Q.5.) Which operation is used to remove an element from a queue? **(0.5 mark)**

0.5

A) peek B) Enqueue ✓ C) Dequeue D) Pop

Q.6.) What happens if you try to pop an element from an empty stack? **(0.5 mark)**

0.5

A) The program enters an infinite loop.
✓ B) A 'Stack Underflow' error occurs.
C) The stack automatically resizes to prevent an error.
D) A 'Stack Overflow' error occurs.

Q.7.) Which of the following is/are linear Data Structures? **(1 mark)**

1

✓ A) Stack ✓ B) Queue C) Tree ✓ D) Linkedlist ✓ E) Array

Q.8.) What is the output of the following program? **(1 mark)**

```
#include <iostream>
using namespace std;
int main()
{
  int arr[] = { 4, 5, 6, 7 };
  int* p = (arr + 2);
```

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```
    cout << *arr + 11;
    return 0;
}
```

A. 16 B. 15 C. 14 D. error

Q.9.) What is the output of the following program? **(1 mark)**

```
#include <iostream>
using namespace std;
int main()
{
    char *ptr;
    char Str[] = "abcdefg";
    ptr = Str;
    ptr += 5;
    cout << ptr;
    return 0;
}
```

A) cdef B) fg C) defg D) abcd